Mohammed Abrar Ahmed

Mohammed Abrar Ahmed | f20221877@hyderabad.bits-pilani.ac.in | +917702108445 | github.com/rarba17

Education

Birla Institute of Technology & Science Pilani, Hyderabad Campus — August 2022 - July 2026
B.E. in Civil Engineering

MS Junior College — Year of Completion: 2021
Class XII: 87%

All Saints High School — Year of Completion: 2019
Class X: 93%

Skills

Subjects / Electives: Foundations of Data Structures & Algorithms, Operating-Systems, Database Systems, Object-Oriented Programming, Artificial Intelligence

Programming: Python, C++, Java, Go

Backend and ML: FastAPI, Docker, LangChain, LangGraph, LangSmith, GPT-4o, llama3.1, OpenAI API, Open-Source LLM's, SQLAlchemy, PostgreSQL, Redis, Alembic, pandas, NumPy

Database & Vector Databases: Postgres, MySQL, SQlite, PineCone, Faiss, Qdrant

Cloud & Tools: Docker, AWS-BedRock, AWS-Lambda, Git, GitHub, MS Excel

Professional Experience

Questt.ai | *SDE Intern* — July 2025 – December 2025

- Designed **RESTful APIs** supporting **50+** concurrent users with thread-based session management, achieving **sub-200ms** average response times.
- Built query processing backend converting business requests into optimized SQL with automatic enhancement and result formatting.
- Orchestrated **multi-agent LLM** workflow processing **500+** requests per day with under **2-second** latency via an async WebSocket pipeline.
- Crafted prompts for document processing achieving **95%** extraction accuracy across **2500+** documents, resulting in an **85%** reduction in manual validation effort.

GMR Varalakshmi Foundation | *Data Analyst Intern* — May 2024 – July 2024

- Analyzed CSR program datasets for GMR Varalakshmi Foundation, converting beneficiary metrics into clear, actionable insights that guided leadership on community-impact strategy
- Built automated dashboards and streamlined reporting workflows, reducing analysis time by **40%** and boosting transparency across multi-state development initiatives

Projects

Financial-Analyst-Agent | *Python, FastApi, Langchain, Postgres, Redis, Streamlit* |

- Architected and developed a full-stack AI financial analyst agent using FastAPI, LangChain, and GPT-4o, enabling natural language querying of P&L data across 10+ global regions, reducing analyst report generation time by **70%**.
- Built secure OTP-based passwordless auth with bcrypt hashing and JWT tokens, reducing login friction by **50%** while supporting 2 role-based tiers.
- Orchestrated multi-agent LLM workflow **(LangChain, LangSmith)** with real-time streaming via async WebSocket pipeline.
- Implemented financial intelligence engine computing **YoY** growth indices, budget vs actual variances, and hierarchical profitability metrics **(EBITDA, EBIT, PAT)**.

Enterprise Document Intelligence Platform | *Python, FastAPI, LangChain, Postgres, PgVector, Celery, Redis, Streamlit* |

- Architected an async **RAG** backend to process PDFs, using **Celery & Redis** to handle high-concurrency workloads without blocking the API.
- Reduced LLM operational costs by **40%** through Redis Semantic Caching, serving repeated queries directly from memory.
- Implemented a unified **PostgreSQL (pgvector)** database to manage high-dimensional vector embeddings alongside relational application data.

AI Onboarding Agent | *Python, FastAPI, PostgresDB, LangGraph, Redis, Streamlit* |

- Built a multi-agent AI onboarding system using **LangGraph** and **GPT-4o** to detect user blockers and deliver real-time nudges via **WebSocket**.
- Engineered a LangGraph state machine orchestrating **4 AI agents**: Diagnosis, Coach, Action Taker, and Escalation with deterministic routing logic.
- Designed an event-driven pipeline with **FastAPI** and **Redis Streams** to ingest user behavior events asynchronously and trigger AI diagnostic workflows.

Leadership And Activities

Core Member - Civil Engineering Association- BITS Hyderabad — 2023-2025

- Co-ordinated and executed CEA's inaugural Colloquium'24, managing four flagship events for 100+ participants.
- Led end-to-end event logistics: venue booking, schedule planning, promotional campaigns (social media posters and captions), and on-site team management.

Founding Member,Paradox Magic Society-BITS Hyderabad — 2023–2025

- Piloted Paradox Magic Society's brand development and strategic vision, attracting 15+ charter members within the first month through targeted marketing initiatives and outreach programs
- Designed and executed 12 monthly workshops and live showcases—featuring guest magicians and peer-led masterclasses—boosting member engagement by over 60 %.